THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Two wires of equal length, one nichrome and the other copper, carry the same current. Which gets hotter? 1

(a) nichrome **(b)** copper **(c)** both equally **(d)** neither

2. For a study with the dependent variable 'height of a plant', the most suitable unit is: 1

(a) kg **(b)** degree C **(c)** s **(d)** cm

3. A student gets a 400x total magnification using a 40x objective lens. The eyepiece lens is: 1

(a) 10x **(b)** 5x **(c)** 20x **(d)** 100x

4. Vaccines are more effective when given: 1

(a) only to adults **(b)** before exposure to disease
(c) never **(d)** after the person dies

5. A scrap yard crane drops the iron when: 1

(a) the iron is heavy **(b)** the current is switched off
(c) the current is on **(d)** the wire is long

6. The time for an object to fall decreases when it is dropped from a lower height. The variables that are related are: 1

(a) colour and time **(b)** shape and colour
(c) height and time **(d)** mass of earth and shape

7. A magnification of 100x means an object of 0.1 mm appears as: 1

(a) 10 mm **(b)** 100 mm **(c)** 0.01 mm **(d)** 1 mm

8. The skin protects us because it is: 1

(a) a physical barrier to germs **(b)** a pathogen
(c) full of sugar **(d)** full of oxygen

9. A student tests materials in a circuit: bulb glows with a coin, key and iron nail; not with eraser, wood and glass. The materials that conduct are: 1

(a) all of them **(b)** eraser, wood and glass
(c) coin, key and iron nail **(d)** none

10. A student gets times of 12.1 s, 11.9 s, 12.0 s for 20 swings. The mean is: 1

(a) 12.1 s **(b)** 12.0 s **(c)** 36.0 s **(d)** 11.9 s

11. A student sees rectangular cells with a clear nucleus but no green bodies. The cells may be from: 1

(a) an onion bulb **(b)** bacteria **(c)** a leaf **(d)** a human cheek

12. Mosquito larvae are found in water that is: 1

(a) boiling **(b)** frozen **(c)** stagnant **(d)** flowing fast

13. A compass needle placed near a battery wire does not deflect. A likely reason is: 1

(a) the switch is open **(b)** the needle is a magnet
(c) the wire is copper **(d)** the cell is new

14. Which of the following is a controlled variable when comparing the cooking of puris at 150 and 180 degrees C? 1

(a) thickness of dough **(b)** oil temperature
(c) puffing time **(d)** colour at the end

15. Which of the following is the correct statement about plant cell vacuoles? 1

(a) They make food
(b) They are large and store water and nutrients
(c) They are absent
(d) They are the control centre

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Differentiate between prokaryotic and eukaryotic cells (one point each). 2

17. How are vaccines different from antibiotics? 2

18. Why does an electromagnet pick up more pins when the number of turns is increased? 2

19. What is an outlier? How should it be handled? 2

20. Differentiate between cytoplasm and nucleus. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how an MCB works. What is the advantage over a fuse? 3

22. Discuss how curiosity helped in discovering a common phenomenon such as the boiling of milk, giving the questions raised. 3

23. Name the cells A, B and C: (A) long, branched cell (B) disc-shaped without a nucleus (C) long and spindle-shaped that contracts. Give function of each. 3

24. Why is the BCG vaccine given to infants? Name the disease. 3

OR

Why does an electric kettle have a thick copper cord but a thin nichrome heating coil? 3

25. The readings of a pendulum for 10 swings are 18.2 s, 18.0 s and 18.4 s. Find the mean time and the time period. 3

OR

An unknown cell shows a nucleus but no wall or chloroplasts. Identify, with two reasons, the type of cell. 3

MINDFORWARD

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A report states 'The more fertiliser we gave the taller the plants got.' Write the variables, the likely hypothesis and ways to test it fairly. **5**

OR

What are the features of viruses? Mention two diseases caused by them and two ways of prevention. **5**

27. A family drinks river water directly. Explain the risks and a solution. **5**

OR

Describe the safety systems in a home electrical circuit: fuse, MCB, earthing, insulation, switches, and say what each protects against. **5**

END OF QUESTION PAPER — Total: 50 Marks

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